

R^n fazo va uning muhim to'plamlari

Reja:

- R^n fazo**
- R^n fazoda ochiq va yopiq to'plamlar**

1^o. R^n fazo. n ta $A_1, A_2, \dots, A_n$ ($n \geq 1$ butun son) to'plamlarning Dekart ko'paytmasi ikkita A va V to'plamlarning Dekart ko'paytmasiga o'xshash ta'riflanadi. Agar $A_1 = A_2 = \dots = A_n = R$ bo'lsa, u holda

$$A_1 \times A_2 \times \dots \times A_n = R \times R \times \dots \times R = \{(x_1, x_2, \dots, x_n) : x_1 \in R, x_2 \in R, \dots, x_n \in R\}$$

bo'ladi. Ushbu

$$\{(x_1, x_2, \dots, x_n) : x_1 \in R, x_2 \in R, \dots, x_n \in R\}$$

to'plam R^n to'plam deb ataladi. R^n to'plamning elementi $(x_1, x_2, \dots, x_n)$ shu to'plam nuqtasi deyiladi va u odatda bitta harf bilan belgilanadi: $x = (x_1, x_2, \dots, x_n)$. Bunda $x_1, x_2, \dots, x_n$ sonlar x nuqtaning mos ravishda birinchi, ikkinchi, ... n -koordinatalari deyiladi.

Agar $x = (x_1, x_2, \dots, x_n) \in R^n$, $y = (y_1, y_2, \dots, y_n) \in R^n$ nuqtalar uchun $x_1 = y_1, x_2 = y_2, \dots, x_n = y_n$ bo'lsa, u holda $x = y$ deb ataladi.

R^n to'plamda ixtiyoriy $x = (x_1, x_2, \dots, x_n)$, $y = (y_1, y_2, \dots, y_n)$ nuqtalarni olaylik.

1-ta'rif. Ushbu

$$\rho(x, y) = \sqrt{(y_1 - x_1)^2 + (y_2 - x_2)^2 + \dots + (y_n - x_n)^2} = \sqrt{\sum_{i=1}^n (y_i - x_i)^2} \quad (12.3)$$

miqdor x va y nuqtalar orasidagi masofa (Evklid masofasi) deb ataladi. Bunday aniqlangan masofa quyidagi xossalarga ega (bunda $\forall x, y, z \in R^n$)

$$1) \rho(x, y) \geq 0 \text{ va } \rho(x, y) = 0 \Leftrightarrow x = y,$$

$$2) \rho(x, y) = \rho(y, x),$$

$$3) \rho(x, z) \leq \rho(x, y) + \rho(y, z)$$

Bu xossalarni isbotlaylik. (12.3) munosabatdan $\rho(x, y)$ miqdorning har doim manfiy emasligini ko'ramiz. Agar $\rho(x, y) = 0$ bo'lsa, unda $y_1 - x_1 = 0$, $y_2 - x_2 = 0, \dots, y_n - x_n = 0$ bo'lib, $x_1 = y_1, x_2 = y_2, \dots, x_n = y_n$ ya'ni $x = y$ bo'ladi. Aksincha $x = y$, ya'ni

$x_1=y_1, x_2=y_2, \dots, x_m=y_m$ bo'lsa, u holda (12.3) dan $\rho(x, y)=0$ bo'lishi kelib chiqadi. Bu esa 1)-xossani isbotlaydi.

(12.3) munosabatdan

$$\begin{aligned}\rho(x, y) &= \sqrt{(y_1-x_1)^2 + (y_2-x_2)^2 + \dots + (y_m-x_m)^2} = \\ &= \sqrt{(x_1-y_1)^2 + (x_2-y_2)^2 + \dots + (x_m-y_m)^2} = \rho(y, x)\end{aligned}$$

bo'ladi.

Masofaning 3)-xossasi ushbu

$$\sqrt{\sum_{i=1}^m (a_i+b_i)^2} \leq \sqrt{\sum_{i=1}^m a_i^2} + \sqrt{\sum_{i=1}^m b_i^2} \quad (12.4)$$

tengsizlikka asosanib isbotlanadi, $a_1, a_2, \dots, a_m; b_1, b_2, \dots, b_m$ ixtiyoriy haqiqiy sonlar. Avvalo shu tengsizlikning to'g'riligini ko'rsataylik. Ravshanki, $\forall x \in R$ uchun

$$\sum_{i=1}^m (a_i x + b_i)^2 \geq 0$$

Bundan, x ga nisbatan kvadrat uchxadning manfiy emasligi

$$\left(\sum_{i=1}^m a_i^2\right)x^2 + \left(2\sum_{i=1}^m a_i b_i\right)x + \sum_{i=1}^m b_i^2 \geq 0$$

kelib chiqadi. Demak, bu kvadrat uchxad ikkita turli haqiqiy ildizga ega bo'lmaydi. Binobarin, uning diskriminanti

$$-\sum_{i=1}^m a_i^2 \sum_{i=1}^m b_i^2 + \left[\sum_{i=1}^m a_i b_i\right]^2 \leq 0$$

bo'lishi kerak. Bundan esa

$$\sum_{i=1}^m a_i b_i \leq \sqrt{\sum_{i=1}^m a_i^2} \sqrt{\sum_{i=1}^m b_i^2}$$

bo'lib,

$$\sum_{i=1}^m a_i^2 + \sum_{i=1}^m b_i^2 + 2\sum_{i=1}^m a_i b_i \leq \left[\sqrt{\sum_{i=1}^m a_i^2}\right]^2 + \left[\sqrt{\sum_{i=1}^m b_i^2}\right]^2 + 2\sqrt{\sum_{i=1}^m a_i^2} \cdot \sqrt{\sum_{i=1}^m b_i^2}$$

bo'ladi. Keyingi tengsizlikdan esa

$$\sqrt{\sum_{i=1}^m (a_i+b_i)^2} \leq \sqrt{\sum_{i=1}^m a_i^2} + \sqrt{\sum_{i=1}^m b_i^2}$$

bo'lishi kelib chiqadi. Odatda (12.4) tengsizlik Koshi-Bunyakovskiy tengsizlik deb ataladi.

Ixtiyoriy $x=(x_1, x_2, \dots, x_m) \in R^m$, $y=(y_1, y_2, \dots, y_m) \in R^m$, $z=(z_1, z_2, \dots, z_m) \in R^m$ nuqtalarni olib, ular orasidagi masofani (12.3) formuladan foydalanib topamiz:

$$\begin{aligned}\rho(x, y) &= \sqrt{\sum_{i=1}^m (y_i - x_i)^2}, \\ \rho(y, z) &= \sqrt{\sum_{i=1}^m (z_i - y_i)^2}, \\ \rho(x, z) &= \sqrt{\sum_{i=1}^m (z_i - x_i)^2}.\end{aligned}\tag{12.5}$$

Endi Koshi-Bunyakovskiy tengsizligi (12.4) da

$$a_i = y_i - x_i, \quad b_i = z_i - y_i \quad (i=1, 2, \dots, m)$$

deb olsak, unda

$$a_i + b_i = z_i - x_i \quad (i=1, 2, \dots, m)$$

bo'lib,

$$\sqrt{\sum_{i=1}^m (z_i - x_i)^2} \leq \sqrt{\sum_{i=1}^m (y_i - x_i)^2} + \sqrt{\sum_{i=1}^m (z_i - y_i)^2}$$

bo'ladi. Yuqoridagi (12.5) munosabatlarni e'tiborga olib, topamiz:

$$\rho(x, z) \leq \rho(x, y) + \rho(y, z).$$

Bu esa 3)-xossani isbotlaydi.

R^m to'plam R^m fazo (m o'lchovli Evklid fazosi) deb ataladi. Endi R^m fazoning ba'zi bir muhim to'plamlarini keltiramiz.

Biror $a=(a_1, a_2, \dots, a_m) \in R^m$ nuqta va $r > 0$ sonni olaylik. Quyidagi

$$\left\{ x=(x_1, x_2, \dots, x_m) \in R^m: (x_1 - a_1)^2 + (x_2 - a_2)^2 + \dots + (x_m - a_m)^2 \leq r^2 \right\}, \tag{12.6}$$

$$\left\{ x=(x_1, x_2, \dots, x_m) \in R^m: (x_1 - a_1)^2 + (x_2 - a_2)^2 + \dots + (x_m - a_m)^2 < r^2 \right\}, \tag{12.7}$$

ya'ni

$$\left\{ x \in R^m: \rho(x, a) \leq r \right\},$$

$$\left\{ x \in R^m: \rho(x, a) < r \right\}$$

to'plamlar mos ravishda shar hamda ochiq shar deb ataladi. Bunda a nuqta shar markazi, r esa shar radiusi deyiladi.

Ushbu

$$\left\{ x = (x_1, x_2, \dots, x_m) \in R^m : (x_1 - a_1)^2 + (x_2 - a_2)^2 + \dots + (x_m - a_m)^2 = r^2 \right\}$$

ya'ni

$$\left\{ x \in R^m : \rho(x, a) = r \right\}$$

to'plam sfera deb ataladi. Bu sfera (12.6) va (12.7) to'plamlarning chegarasi bo'ladi.

Ushbu

$$\left\{ x = (x_1, x_2, \dots, x_m) \in R^m : a_1 \leq x_1 \leq b_1, a_2 \leq x_2 \leq b_2, \dots, a_m \leq x_m \leq b_m \right\},$$

$$\left\{ x = (x_1, x_2, \dots, x_m) \in R^m : a_1 < x_1 < b_1, a_2 < x_2 < b_2, \dots, a_m < x_m < b_m \right\}$$

to'plamlar (bunda $a_1, a_2, \dots, a_m; b_1, b_2, \dots, b_m$ haqiqiy sonlar) mos ravishda parallelepiped hamda ochiq parallelepiped deb ataladi.

Ushbu

$$\left\{ x = (x_1, x_2, \dots, x_m) \in R^m : x_1 \geq 0, x_2 \geq 0, \dots, x_m \geq 0, x_1 + x_2 + \dots + x_m \leq h \right\}$$

to'plam (m o'lchovli) simpleks deb ataladi, bunda h - musbat son.

Yuqorida keltirilgan to'plamlar tez-tez ishlatilib turiladi. Ular yordamida muhim tushunchalar, jumladan atrof tushunchasi ta'riflanadi.

2^o. R^m fazoda ochiq va yopiq to'plamlar. Biror $x^0 = (x_1^0, x_2^0, \dots, x_m^0) \in R^m$ nuqta hamda $\varepsilon > 0$ sonni olaylik.

2-ta'rif. Markazi x^0 nuqtada, radiusi $\varepsilon > 0$ ga teng bo'lgan ochiq shar x^0 nuqtaning sferik atrofi (ε atrofi) deyiladi va $U_\varepsilon(x^0)$ kabi belgilanadi:

$$U_\varepsilon(x^0) = \left\{ x \in R^m : \rho(x, x^0) < \varepsilon \right\}.$$

Nuqtaning boshqacha atrofi tushunchasini ham kiritishimiz mumkin.

3-ta'rif. Ushbu

$$\left\{ x = (x_1, x_2, \dots, x_m) \in R^m : x_1^0 - \delta_1 < x_1 < x_1^0 + \delta_1, \right. \\ \left. x_2^0 - \delta_2 < x_2 < x_2^0 + \delta_2, \dots, x_m^0 - \delta_m < x_m < x_m^0 + \delta_m \right\} \quad (12.8)$$

ochiq parallelepiped x^0 nuqtaning parallelepipedial atrofi deb ataladi va $\tilde{U}_{\delta_1 \delta_2 \dots \delta_m}(x^0)$ kabi belgilanadi.

Xususan $\delta_1 = \delta_2 = \dots = \delta_m = \delta$ bo'lsa, (12.8) ochiq parallelepiped kubga aylanadi va uni $\tilde{U}_\delta(x^0)$ kabi belgilanadi.

1-lemma. $x^0 \in R^m$ nuqtaning har qanday $U_\varepsilon(x^0)$ sferik atrofi olinganda ham har doim x^0 nuqtaning shunday $\tilde{U}_{\delta_1 \delta_2 \dots \delta_m}(x^0)$ parallelepipedial atrofi mavjudki, bunda

$$\tilde{U}_{\delta_1 \delta_2 \dots \delta_m}(x^0) \subset U_\varepsilon(x^0)$$

bo'ladi.

Shuningdek, x^0 nuqtaning har qanday $\tilde{U}_{\delta_1 \delta_2 \dots \delta_m}(x^0)$ parallelepipedial atrofi olinganda ham har doim shu nuqtaning shunday $U_\varepsilon(x^0)$ sferik atrofi mavjudki, bunda

$$U_\varepsilon(x^0) \subset \tilde{U}_{\delta_1 \delta_2 \dots \delta_m}(x^0)$$

bo'ladi.

$x^0 \in R^m$ nuqtaning sferik atrofi

$$U_\varepsilon(x^0) = \{x \in R^m : \rho(x, x^0) < \varepsilon\}$$

berilgan bo'lsin. Bundagi $\varepsilon > 0$ songa ko'ra $\delta < \frac{\varepsilon}{\sqrt{m}}$ tengsizlikni qanoatlantiruvchi $\delta > 0$ sonni olamiz. So'ng x^0 nuqtaning ushbu

$$\tilde{U}_\delta(x^0) = \{x = (x_1, x_2, \dots, x_m) \in R^m : x_1^0 - \delta < x_1 < x_1^0 + \delta, \dots, x_m^0 - \delta < x_m < x_m^0 + \delta\}$$

parallelepipedial atrofni tuzamiz.

Aytaylik, $x \in \tilde{U}_\varepsilon(x^0)$ bo'lsin. Unda $|x_i - x_i^0| < \delta \quad (i=1, 2, \dots, m)$ bo'lib,

$$\sqrt{\sum_{i=1}^m (x_i - x_i^0)^2} < \sqrt{\sum_{i=1}^m \delta^2} = \delta \sqrt{m}$$

bo'ladi. Yuqoridagi $\delta < \frac{\varepsilon}{\sqrt{m}}$ tengsizlikni e'tiborga olib topamiz:

$$\rho(x, x^0) = \sqrt{\sum_{i=1}^m (x_i - x_i^0)^2} < \varepsilon$$

Demak, $\rho(x, x^0) < \varepsilon$. Bu esa $x \in U_\varepsilon(x^0)$ ekanini bildiradi. Shunday qilib,

$$\forall x \in \tilde{U}_\delta(x^0) \Rightarrow x \in U_\varepsilon(x^0)$$

ya'ni

$$\tilde{U}_\delta(x^0) \subset U_\varepsilon(x^0)$$

bo'ladi.

$x^0 \in R^m$ nuqtaning

$$\tilde{U}_{\delta_1 \delta_2 \dots \delta_m}(x^0) = \text{parallelepiped}$$

parallelepiped atrofi berilgan bo'lsin. Unda

$$\varepsilon = \min\{\delta_1, \delta_2, \dots, \delta_m\}$$

ni olib x^0 nuqtaning sferik atrofi

$$U_\varepsilon(x^0) = \{x \in R^m : \rho(x, x^0) < \varepsilon\}$$

ni tuzamiz.

Aytaylik $x \in U_\varepsilon(x^0)$ bo'lsin. U holda

$$\rho(x, x^0) = \sqrt{\sum_{i=1}^m (x_i - x_i^0)^2} < \varepsilon \leq \delta_i \quad (i=1, 2, \dots, m)$$

bo'ladi. Demak,

$$|x_i - x_i^0| \leq \sqrt{\sum_{i=1}^m (x_i - x_i^0)^2} < \delta_i \quad (i=1, 2, \dots, m)$$

Bundan esa $x \in \tilde{U}_{\delta_1 \delta_2 \dots \delta_m}(x^0)$ bo'lishi kelib chiqadi. Shunday qilib,

$$\forall x \in U_\varepsilon(x^0) \Rightarrow x \in \tilde{U}_{\delta_1 \delta_2 \dots \delta_m}(x^0)$$

ya'ni

$$U_\varepsilon(x^0) \subset \tilde{U}_{\delta_1 \delta_2 \dots \delta_m}(x^0)$$

bo'ladi. □

Aytaylik, R^m fazo va G to'plam berilgan bo'lsin: $G \subset R^m$

4-ta'rif. Agar $x^0 \in G$ nuqtaning shunday $U_\delta(x^0)$ atrofi ($\delta > 0$) topilsaki,

$$U_\delta(x^0) \subset G$$

bo'lsa, x^0 nuqta G to'plamning ichki nuqtasi deyiladi.

5-ta'rif. To'plamning har bir nuqtasi uning ichki nuqtasi bo'lsa, u ochiq to'plam deb ataladi.

12.1-misol. R^m fazodagi ochiq shar ochiq to'plam bo'lishi isbotlansin.

□ Aytaylik,

$$A = \{x \in R^m : \rho(x, x^0) < r\}$$

R^m fazodagi biror ochiq shar bo'lsin. Ravshanki, $z \in A \Rightarrow \rho(z, x^0) < r$.

Ushbu $r_1 = r - \rho(z, x^0)$ sonni olib, quyidagi

$$B = \{x \in R^m : \rho(x, x^0) < r_1\}$$

sharni qaraymiz. $\forall y \in B$ uchun,

$$\rho(y, x^0) \leq \rho(y, z) + \rho(z, x^0) < r_1 + \rho(z, x^0)$$

munosabatga ko'ra

$$B \subset A$$

bo'ladi. □

R^m fazoda biror F to'plam va biror x^0 nuqta berilgan bo'lsin:

6-ta'rif. Agar

$$\forall r > 0, \exists x \in F, x \neq x^0 : x \in \{x \in R^m : \rho(x, x^0) < r\}$$

bo'lsa, x^0 nuqta F to'plamning limit nuqtasi deyiladi.

Masalan, ushbu

$$A = \{x \in R^m: \rho(x, x^0) \leq r\}$$

sharning barcha nuqtalari uning limit nuqtalari bo'ladi.

7-ta'rif. $F \subset R^m$ to'plamning barcha limit nuqtalari shu to'plamga tegishli bo'lsa, F yopiq to'plam deb ataladi.

Masalan,

$$E = \{x \in R^m: \rho(x, x^0) \leq r\}$$

yopiq to'plam bo'ladi.

Shuni ta'kidlash lozimki, ochiq va yopiq to'plamlar ta'riflarini qanoatlantirmaydigan to'plamlar ham ko'pdir.

Biror $M \subset R^m$ to'plamni qaraylik. Ravshanki, R^m ayirma M to'plamni R^m to'plamga to'ldiruvchi to'plam bo'ladi. (qaralsin I-qism, I-bob, I-§).

8-ta'rif. Agar $x^0 (x^0 \in R^m)$ nuqtaning istalgan $U_\varepsilon(x^0)$ atrofida ham M to'plamning, ham R^m to'plamning nuqtalari bo'lsa, x^0 nuqta M to'plamning chegaraviy nuqtasi deb ataladi. M to'plamning barcha chegaraviy nuqtalaridan iborat to'plam M to'plamning chegarasi deyiladi va uni odatda $\partial(M)$ kabi belgilanadi.

Bu tushuncha yordamida yopiq to'plamni quyidagicha ham ta'riflash mumkin.

9-ta'rif. Agar $F (F \subset R^m)$ to'plamning chegarasi shu to'plamga tegishli, ya'ni $\partial(F) \subset F$ bo'lsa, F yopiq to'plam deb ataladi.

Yopiq to'plamning yuqorida keltirilgan 12.7- va 12.9- ta'riflari ekvivalent ta'riflardir.

Biror $M \subset R^m$ to'plam berilgan bo'lsin.

10-ta'rif. Agar R^m fazoda shunday shar

$$U^0 = \{x \in R^m: \rho(x, 0) < r\} \quad (0 = (0, 0, \dots, 0))$$

topilsaki, $M \subset U^0$ bo'lsa, M chegaralangan to'plam deb ataladi.

Faraz qilaylik, $x_1(t), x_2(t), \dots, x_m(t)$ funksiyalarning har biri $[a, b]$ segmentda uzluksiz bo'lsin.

Ushbu

$$\{x_1(t), x_2(t), \dots, x_m(t)\} \quad (a \leq t \leq b) \quad (12.9)$$

sistema yoki nuqtalar to'plami R^m fazoda egri chiziq deb ataladi. Xususan,

$$x_1 = \alpha_1 t + \beta_1, \quad x_2 = \alpha_2 t + \beta_2, \dots, \quad x_m = \alpha_m t + \beta_m$$

($-\infty < t < +\infty$, $\alpha_1, \alpha_2, \dots, \alpha_m$; $\beta_1, \beta_2, \dots, \beta_m$ haqiqiy sonlar va $\alpha_1, \alpha_2, \dots, \alpha_m$ larning xech bo'lmaganda bittasi nolga teng emas) bo'lganda (12.9) sistema R^m fazoda to'g'ri chiziq deyiladi.

R^m fazoda ixtiyoriy ikkita $x' = (x'_1, x'_2, \dots, x'_m)$ va $x'' = (x''_1, x''_2, \dots, x''_m)$ nuqtani olaylik. Bu nuqtalar orqali o'tuvchi to'g'ri chiziq quyidagi

$$\left\{ \left(x'_1 + t(x''_1 - x'_1), \quad x'_2 + t(x''_2 - x'_2), \quad \dots, \quad x'_m + t(x''_m - x'_m) \right) \right\} \quad (-\infty < t < +\infty) \quad (12.10)$$

sistemi bilan ifodalanadi. Bunda $t=0$ va $t=1$ bo'lganda R^m fazoning mos ravishda x' va x'' nuqtalari hosil bo'lib, $0 \leq t \leq 1$ bo'lganda (12.10) sistema R^m fazoda x' va x'' nuqtalarni birlashtiruvchi to'g'ri chiziq kesmasi bo'ladi.

R^m fazoda chekli sondagi to'g'ri chiziq kesmalarni birin-ketin birlashtirishdan tashkil topgan chiziq, siniq chiziq deb ataladi.

$M \subset R^m$ to'plam berilgan bo'lsin.

11-ta'rif. Agar M to'plamning ixtiyoriy ikki nuqtasini birlashtiruvchi shunday siniq chiziq topilsaki, u M to'plamga tegishli bo'lsa, M bog'lamli to'plam deb ataladi.

12-ta'rif. Agar $M \subset R^m$ to'plam ochiq hamda bog'lamli to'plam bo'lsa, u soha deb ataladi.

R^m fazoda ochiq parallelepiped, ochiq shar, ochiq simplekslar fazodagi sohalar bo'ladi.

Adabiyotlar

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